

I. Ulas Ayyilmaz

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EDUCATION

Bernstein Center & Technische Universität Berlin, Berlin, Germany

Expected 2027

MSc Computational Neuroscience

Pomona College, Claremont, CA

Dec 2024

BA Computer Science | GPA: 3.8

Coursework: Computational Statistics, Machine Learning, Managing Complex Systems, Algorithms, Affective Neuroscience, Life Span Development, Artificial & Biological Vision, Sports Psychology

CONFERENCE PRESENTATIONS

Talks

Ulas I. Ayyilmaz. (2024), “Excitatory and inhibitory neurons exhibit distinct roles for task learning, temporal scaling, and working memory in recurrent spiking neural network models of neocortex”, Spiking Neural Networks as Universal Function Approximators Online Conference (SNUFA).

Posters

Ulas I. Ayyilmaz. (2024), “Excitatory and inhibitory neurons exhibit distinct roles for task learning, temporal scaling, and working memory in recurrent spiking neural network models of neocortex”, Bernstein Conference, Frankfurt, Germany.

Ulas I. Ayyilmaz. (2024), “Investigating the Effect of Shared Gaze on Multi-Person Teams”, Harvey Mudd Computer Science REU Symposium, Claremont, CA, USA.

RESEARCH EXPERIENCE

Claremont McKenna College, CA, USA

Lab Manager, Enkavi Neuroimaging Lab

Jan 2025 – Jun 2025

- Streamlined quality checks of fMRI scans by building a web page that automates visualization of OpenNeuro data
- Built an end-to-end pipeline for data acquisition, fmripred execution, and storage space optimization across hundreds of subject scans on an HPC cluster
- Built and deployed the lab’s official website from scratch using Html, Css, and Javascript

Pomona College, CA, USA

Research Assistant, Human-Computer Interaction Lab

Jun 2024 – Oct 2024

- Designed an experimental process to assess how partner gaze visualizations influence remote group collaborations
- Conducted 90-minute experimental sessions with 26 distinct groups over five months
- Conducted post-study interviews, documented participant feedback, and cataloged multimodal data

Research Assistant, Zhu Computational Neuroscience Lab Jan 2024 – April 2025

- Simulated interneuron diversity (PV, SST, and 5HT) in the cortex by building biologically plausible Recurrent Spiking Neural Network (RSNN) models on Pytorch
- Trained models on various temporal tasks including tasks that require synchronization continuation and robustness to noise
- Analyzed unique roles of interneuron types in temporal scaling, phase-tuning, signal amplification, and disinhibition, by examining changes in connectivity, spiking behavior and task accuracy
- Mentored lab members on computational tools including Pytorch and HPC clusters

TEACHING EXPERIENCE

Pomona College, CA, USA

Teaching Assistant, Electronics (Phys 128) Jan – May 2022

Teaching Assistant, Introduction to Computer Science (CSCI 51) Sep – Dec 2021

Amerikan Kultur Dil Okullari, Izmir, Turkey

English B1-Level Instructor Jun – Jul 2023

PROJECTS

Predicting Mirror Neuron Activity in Motor Cortex by Training Convolutional Neural Networks on EEG Signals Sep – Dec 2023

Predicting Intrinsically Disordered Proteins with Random Forest Classifiers based on Amino Acid Sequences and Genetic Characteristics Jan – May 2022

INDUSTRY EXPERIENCE

Marriott International, Izmir, Turkey

Front Desk Agent May 2023 – Aug 2023

- Managed check-in, check-out processes; handling cash and valuables
- Wrote a novel 10-page operations manual to streamline front desk training

Digiboard, CA, USA

Sep 2022 – Mar 2023

Product Manager Intern

- Developed and deployed a start-up mobile app that provided a platform for 8,000+ college students to organize, share and submit events
- Communicated technical updates to marketing and finance teams during weekly presentations
- Interviewed 50+ students and local businesses, and extracted insights to understand user pain points

LEADERSHIP EXPERIENCE

Pomona College, CA, USA

Founder & Captain, All Gender Volleyball Club (*220+ members as of 2024*) Aug 2022 – Dec 2024

Peer Advisor, Career Development Office Aug 2022 – May 2023

President, Turkish Student Group (*Led a \$16,000+ donation campaign*)

Aug 2021 – May 2023

AWARDS

Ted Gleason Award, Pomona College

May 2024

Awarded to a graduating senior demonstrating exceptional community support and leadership.

REFERENCES

A. Zeynep Enkavi, PhD

Assistant Professor of Integrated Sciences: Neuroscience

Claremont McKenna College

zenkavi@cmc.edu

Yuqing Zhu, PhD

Visiting Assistant Professor of Neuroscience

Pomona College

Yuqing.Zhu@pomona.edu

Alexandra Papoutsaki, PhD

Associate Professor of Computer Science

Pomona College

Alexandra.Papoutsaki@pomona.edu

SKILLS & INTERESTS

Technical: Python, Unix Shell, R, FSL, SLURM, Canva, HTML, CSS, JavaScript, Java, C++

Languages: English (C2), Turkish (Native), German(A2), Spanish (A2)

Interest: Creative Writing, Volleyball Coaching, Turkish Folk Dance, Piano, Swimming